

EC4219: Software Engineering

Lecture 4 — First-Order Theories

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Review: First-Order Logic

- FOL is an extension of PL with quantifiers and nonlogical symbols. A first-order logic consists of
 - ▶ variables,
 - ▶ logical symbols (boolean connectives and quantifiers), and
 - ▶ nonlogical symbols (constant-, function-, predicate symbols)
- The semantics is determined by an interpretation $I : (D_I, \alpha_I)$.
 - ▶ A **domain** D_I is a nonempty set of values such as integers.
 - ▶ An **assignment** α_I is a mapping for free variables and nonlogical symbols.

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Q. What about the following formula?

$$\exists x.2x = 7$$

In FOL formulas, nonlogical symbols are **uninterpreted!**
(i.e., can be assigned any meaning)

Necessaity of First-Order Theories

- In practice, we are interested in a specific class of interpretations. That is, we have **fixed meanings** for some nonlogical symbols!
 - ▶ Given $F : \exists x.x + 0 = 1$, we usually interpret $+$ as $+\mathbb{Z}$.
- First-order logic is rather a general framework for building specific logic, called **First-order theories** that impose restrictions on nonlogical symbols and their meanings.
 - ▶ In the theory of integers ($T_{\mathbb{Z}}$), the meaning of $+$ is fixed as $+\mathbb{Z}$.
- Q.How to restrict interpretations (meanings)?
 - A. By providing a set of axioms. That is, we consider interpretations that satisfy the axioms only.

First-Order Theories

A first-order theory T is defined by the two components.

- **Signature:** A set of nonlogical symbols (constant-, function-, predicate symbols). Given a signature Σ , a Σ -formula is the formula constructed with nonlogical symbols of Σ .
- **Axioms:** A set of closed FOL formulas whose nonlogical symbols are from Σ .

Signature restricts the syntax, and axioms restrict the interpretations.

Basic Terminologies

- An interpretation I is called a **T -interpretation**, if I satisfies all axioms $\mathcal{A}$ of T ,

$$I \models A \text{ for every } A \in \mathcal{A}$$

- A Σ -formula F is **satisfiable in T** or **T -satisfiable**, if there is a T -interpretation that satisfies F .

$$I \models F \text{ for some } T\text{-interpretation } I$$

- A Σ -formula F is **valid in T** or **T -valid**, if every T -interpretation satisfies F .

$$I \models F \text{ for every } T\text{-interpretation } I \text{ (sometimes denoted } T \models F)$$

- A theory T is **complete**, if for every closed Σ -formula F , $T \models F$ or $T \models \neg F$.
- A theory T is **decidable** if $T \models F$ (checking T -validity) is decidable for every Σ -formula F .
 - ▶ There is an algorithm that always terminates with “yes” if F is T -valid or with “no” if F is T -invalid.

Terminologies (cont'd)

A theory restricts only the nonlogical symbols. Restrictions on the logical symbols or the grammar are done by defining **fragments** of the logic. Two popular fragments:

- **Quantifier-free fragment:** the set of Σ -formulas without quantifiers.
- **Conjunctive fragment:** the set of formulas where the only boolean connective that is allowed is conjunction.

Many first-order theories are undecidable while their quantifier-free fragments are decidable. In practice, we are mostly interested in the satisfiability problem of the quantifier-free fragment of first-order theories.

Plan

In the remainder of this lecture, we will explore commonly-used first-order theories.

- The theory of equality T_E
- Peano Arithmetic T_{PA}
- Presburger Arithmetic $T_{\mathbb{N}}$
- The theory of Reals $T_{\mathbb{R}}$ and Rationals $T_{\mathbb{Q}}$.
- The theory of Arrays T_A

Theory of Equality

The theory of equality T_E is the simplest and most widely-used first-order theory. Its signature

$$\Sigma_E : \{=, a, b, c, \dots, f, g, h, \dots, p, q, r, \dots\}$$

consists of

- $=$ (equality), a binary predicate, and
- all constants, function- and predicate symbols.

Equality $=$ is an **interpreted** predicate symbol; its meaning will be defined via the axioms. The others are **uninterpreted** since functions, predicates, and constants are left unspecified.

Axioms of the Theory of Equality

- ① Reflexivity: $\forall x. x = x$
- ② Symmetry: $\forall x, y. x = y \implies y = x$
- ③ Transitivity: $\forall x, y, z. x = y \wedge y = z \implies x = z$
- ④ Function congruence (consistency): for each positive integer n and n -ary function symbol f ,

$$\forall \vec{x}, \vec{y}. \left(\bigwedge_{i=1}^n x_i = y_i \right) \rightarrow f(\vec{x}) = f(\vec{y}).$$

where $\vec{x} = x_1 \cdots x_n$ and $\vec{y} = y_1 \cdots y_n$.

- ⑤ Predicate congruence (consistency): for each positive integer n and n -ary predicate symbol p ,

$$\forall \vec{x}, \vec{y}. \left(\bigwedge_{i=1}^n x_i = y_i \right) \rightarrow p(\vec{x}) = p(\vec{y}).$$

cf) 4 and 5 are axiom schemata; f and p should be instantiated to concrete function- and predicate symbols.

Example 1: Theory of Equality

To prove that

$$F : \forall x, y, z. ((x = y) \wedge \neg(y = z)) \rightarrow \neg(x = z)$$

is T_E -valid, assume otherwise to derive a contradiction.

- | | | |
|-----|--------------------------------------|--------------------|
| 1. | $I \not\models F$ | assumption |
| 2. | $I \models x = y \wedge \neg(y = z)$ | 1, $\rightarrow$ |
| 3. | $I \not\models \neg(x = z)$ | 1, $\rightarrow$ |
| 4. | $I \models x = y$ | 2, $\wedge$ |
| 5. | $I \models \neg(y = z)$ | 2, $\wedge$ |
| 6. | $I \models x = z$ | 3, $\neg$ |
| 7. | $I \models y = x$ | 4, reflexivity |
| 8. | $I \models y = z$ | 6, 7, transitivity |
| 9. | $I \not\models y = z$ | 5, $\neg$ |
| 10. | $I \models \perp$ | 8, 9 |

Example 2: Theory of Equality

Prove that F is T_E -valid.

$$F : a = b \wedge b = c \rightarrow g(f(a), b) = g(f(c), a)$$

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- | | | |
|-----|---|---------------------------|
| 1. | $I \not\models F$ | assumption |
| 2. | $I \models a = b \wedge b = c$ | 1, $\rightarrow$ |
| 3. | $I \not\models g(f(a), b) = g(f(c), a)$ | 1, $\rightarrow$ |
| 4. | $I \models a = b$ | 2, $\wedge$ |
| 5. | $I \models b = c$ | 2, $\wedge$ |
| 6. | $I \models a = c$ | 4, 5, transitivity |
| 7. | $I \models f(a) = f(c)$ | 6, function congruence |
| 8. | $I \models b = a$ | 4, symmetry |
| 9. | $I \models g(f(a), b) = g(f(c), a)$ | 7, 8, function congruence |
| 10. | $I \models \perp$ | 3, 9 |

Decidability

Like the full first-order logic, T_E -validity is undecidable. However, there exists an efficient decision procedure for its quantifier-free fragment.¹

¹see Chap.9 in “The Calculus of Computation: Decision Procedures with Applications to Verification”

Uninterpreted Functions

- In T_E , function symbols are uninterpreted since the axioms do not assign meaning to them other than in the context of equality.
- The only thing we know about them is that they are functions.

Use of Uninterpreted Functions

A main application of uninterpreted functions is to abstract complex formulas that are otherwise difficult to automatically reason about.

- Given a formula F , treating a function symbol f as uninterpreted makes the formula weaker; we ignore the semantics of f (except for congruence with respect to equality).
- Let φ^{UF} be the formula derived from φ by replacing some interpreted functions with uninterpreted ones. Then,

$$\models \varphi^{UF} \implies \models \varphi.$$

Note that the converse is not true!

- φ^{UF} is an approximation of φ such that if φ^{UF} is valid so is φ . But φ^{UF} may fail to be valid even though φ is.

Uninterpreted functions simplify proofs. Uninterpreted functions enable to reason about systems while ignoring the semantics of irrelevant parts.

Example: Use of Uninterpreted Functions

How to prove the equivalence of the two C functions?

```
1 int power3 (int in) {  
2     int i, out;  
3     out = in;  
4     for (i=0; i<2; i++)  
5         out = out * in;  
6     return out;  
7 }
```

```
1 int mypower3 (int in) {  
2     int out;  
3     out = (in * in) * in;  
4     return out;  
5 }
```

We can prove the equivalence by translating the programs into formulas

$$\varphi_a : out_0 = in \wedge out_1 = out_0 * in \wedge out_2 = out_1 * in$$

$$\varphi_b : out = (in * in) * in$$

and checking the validity of the following formula:

$$\varphi_a \wedge \varphi_b \rightarrow out_2 = out$$

Example: Use of Uninterpreted Functions (cont'd)

Deciding formulas with multiplication is generally hard. Replacing the multiplication symbol with uninterpreted functions can aid the problem.

$$\begin{aligned}\varphi_a^{UF} &: out_0 = in \wedge out_1 = G(out_0, in) \wedge out_2 = G(out_1, in) \\ \varphi_b^{UF} &: out = G(G(in, in), in)\end{aligned}$$

The following abstract formula is valid and so is the original formula.

$$\varphi_a^{UF} \wedge \varphi_b^{UF} \rightarrow out_2 = out$$

Theory of Peano Arithmetic

A theory for natural numbers. The theory of Peano arithmetic T_{PA} has the signature

$$\Sigma_{PA} : \{0, 1, +, \cdot, =\}$$

where

- 0 and 1 are constants,
- $+$ (addition) and $\cdot$ (multiplication) are binary functions, and
- $=$ (equality) is a binary predicate.

Axioms: Theory of Peano Arithmetic

The axioms of T_{PA} :

- ① Zero: $\forall x. \neg(x + 1 = 0)$
- ② Successor: $\forall x, y. x + 1 = y + 1 \rightarrow x = y$
- ③ Plus zero: $\forall x. x + 0 = x$
- ④ Plus successor: $\forall x, y. x + (y + 1) = (x + y) + 1$
- ⑤ Times zero: $\forall x. x \cdot 0 = 0$
- ⑥ Times successor: $\forall x, y. x \cdot (y + 1) = x \cdot y + x$
- ⑦ Induction (axiom schema):
 $F[0] \wedge (\forall x. F[x] \rightarrow F[x + 1]) \rightarrow \forall x. F[x]$ (for every Σ_{PA} -formula F with exactly one free variable)

Example: T_{PA} formulas

- The formula $3x + 5 = 2y$ can be written as

$$(1 + 1 + 1) \cdot x + 1 + 1 + 1 + 1 + 1 = (1 + 1) \cdot y$$

- The inequality $3x + 5 > 2y$ can be expressed by

$$\exists z. z \neq 0 \wedge 3x + 5 = 2y + z$$

where $\neg(z = 0) \equiv z \neq 0$.

- Every formula of the set

$$\{\forall x, y, z. x \neq 0 \wedge y \neq 0 \wedge z \neq 0 \rightarrow x^n + y^n \neq z^n \mid n > 2 \wedge n \in \mathbb{Z}\}$$

is T_{PA} -valid (Fermat's Last Theorem).

Decidability and Completeness of T_{PA}

- T_{PA} is neither complete nor decidable.
- Even undecidable is its quantifier-free fragment.
- A fragment of T_{PA} , called Presburger arithmetic, is both complete and decidable.

Axioms: Theory of Presburger Arithmetic

A restriction that does not allow multiplication. The theory has a signature

$$\Sigma_{\mathbb{N}} : \{0, 1, +, =\}$$

and axioms:

- ① Zero: $\forall x. \neg(x + 1 = 0)$
- ② Successor: $\forall x, y. x + 1 = y + 1 \rightarrow x = y$
- ③ Plus zero: $\forall x. x + 0 = x$
- ④ Plus successor: $\forall x, y. x + (y + 1) = (x + y) + 1$
- ⑤ Induction: $F[0] \wedge (\forall x. F[x] \rightarrow F[x + 1]) \rightarrow \forall x. F[x]$

Theory of Integers

- Although integer reasoning can be done with natural numbers, it is convenient to have a theory of integers.
- The theory of integers $T_{\mathbb{Z}}$ (with linear arithmetic) has signatures

$$\Sigma_{\mathbb{Z}} : \{\dots, -1, 0, 1, \dots, -3\cdot, -2\cdot, 2\cdot, 3\cdot, \dots, +, -, =, >\}$$

- $T_{\mathbb{Z}}$ is no more expressive but more convenient than Presburger arithmetic ($T_{\mathbb{N}}$).
- $T_{\mathbb{Z}}$ is both complete and decidable, and one of the most widely used theories.

cf) Integer Reasoning with Natural Number

- Integer reasoning can be performed with natural number reasoning: formulas over all integers $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$ can be encoded as $\Sigma_{\mathbb{N}}$ -formulas.
- Idea: replace integer variables with the difference of variables of natural-numbers. For example, consider the formula

$$F_0 : \forall w, x. \exists y, z. x + 2y - z > -3w$$

- 1 Introduce two variables, v_p and v_n , for each variable v of F_0 :

$$F_1 : \forall w_p, w_n, x_p, x_n. \exists y_p, y_n, z_p, z_n. \\ (x_p - x_n) + 2(y_p - y_n) - (z_p - z_n) > -3(w_p - w_n)$$

- 2 Move negated terms to the other side of the inequality.

$$F_2 : \forall w_p, w_n, x_p, x_n. \exists y_p, y_n, z_p, z_n. \\ x_p + 2y_p + z_n + 3w_p > x_n + 2y_n + z_p + 3w_n$$

F_2 is $T_{\mathbb{N}}$ -valid precisely when F_0 is valid in the integer interpretation.

Theories of Reals and Rationals

The theory of reals $T_{\mathbb{R}}$ has the signature

$$\Sigma_{\mathbb{R}} : \{0, 1, +, -, \cdot, =, \geq\}.$$

The theory of rationals $T_{\mathbb{Q}}$ has the signature

$$\Sigma_{\mathbb{Q}} : \{0, 1, +, -, =, \geq\}.$$

$T_{\mathbb{R}}$ and $T_{\mathbb{Q}}$ have complex axioms (see Chapter 3 of the textbook).

Theory of Arrays

The theory of arrays T_A has the signature

$$\Sigma_A : \{\cdot[\cdot], \cdot\langle\cdot\triangleleft\cdot\rangle, =\}$$

where

- $a[i]$ represents the value of array a at position i (binary function).
- $a\langle i\triangleleft v\rangle$ represents the modified array a in which position i has the value v (ternary function).
- $=$ is the equality predicate.

The axioms of T_A :

- 1 the axioms of reflexivity, symmetry, and transitivity of T_E
- 2 (array congruence) $\forall a, i, j. i = j \rightarrow a[i] = a[j]$
- 3 (read-over-write 1) $\forall a, i, j. i = j \rightarrow a\langle i\triangleleft v\rangle[j] = v$
- 4 (read-over-write 2) $\forall a, i, j. i \neq j \rightarrow a\langle i\triangleleft v\rangle[j] = a[j]$

Example: Theory of Arrays

Determine the validity of the formula:

$$F : a[i] = e \rightarrow \forall j. a\langle i \triangleleft e \rangle[j] = a[j]$$

- | | | |
|-----|--|-----------------------------------|
| 1. | $I \not\models F$ | assumption |
| 2. | $I \models a[i] = e$ | 1, $\rightarrow$ |
| 3. | $I \not\models \forall j. a\langle i \triangleleft e \rangle[j] = a[j]$ | 1, $\rightarrow$ |
| 4. | $I_1 : I \triangleleft \{j \mapsto v\} \not\models a\langle i \triangleleft e \rangle[j] = a[j]$ | 3, $\forall$, for some $v \in D$ |
| 5. | $I_1 \models a\langle i \triangleleft e \rangle[j] \neq a[j]$ | 4, $\neg$ |
| 6. | $I_1 \models i = j$ | 5, read-over-write 2 |
| 7. | $I_1 \models a[i] = a[j]$ | 6, array congruence |
| 8. | $I_1 \models a\langle i \triangleleft e \rangle[j] = e$ | 6, read-over-write 1 |
| 9. | $I_1 \models a\langle i \triangleleft e \rangle[j] = a[j]$ | 2, 7, 8, transitivity |
| 10. | $I_1 \models \perp$ | |

Decidability of First-order Theories

Theory	Description	Full	QFF
T_E	equality	no	yes
T_{PA}	Peano arithmetic	no	no
$T_{\mathbb{N}}$	Presburger arithmetic	yes	yes
$T_{\mathbb{Z}}$	linear integers	yes	yes
$T_{\mathbb{R}}$	reals (with $\cdot$)	yes	yes
$T_{\mathbb{Q}}$	rational (without $\cdot$)	yes	yes
T_{RDS}	recursive data structures	no	yes
T_A	arrays	no	yes
$T_A^=$	arrays with extensionality	no	yes

cf) Combining Theories

- In practice, the formulas we check for satisfiability or validity span multiple theories.
 - ▶ For example, in program verification, we want to prove properties about a list of integers or an array of integers.
- Nelson and Oppen presented a general method for combining quantifier-free fragments of first-order theories.
- Given T_1 and T_2 such that $\Sigma_1 \cap \Sigma_2 = \{=\}$, the combined theory $T_1 \cup T_2$ has the signature $\Sigma_1 \cup \Sigma_2$ and axioms $A_1 \cup A_2$. Nelson and Oppen showed that if
 - ▶ satisfiability in the quantifier-free fragments of T_1 is decidable
 - ▶ satisfiability in the quantifier-free fragments of T_2 is decidable
 - ▶ and certain conditions are met

then satisfiability in the quantifier-free fragment of $T_1 \cup T_2$ is decidable.

- Furthermore, if the decision procedures for T_1 and T_2 are in P (resp., NP), then the combined decision procedure for $T_1 \cup T_2$ is in P (resp., NP).

Summary

- FOL is an extension of PL with quantifiers and nonlogical symbols (constant-, function-, and predicate symbols).
- In FOL formulas, nonlogical symbols are uninterpreted!
 - ▶ $\exists x.x + 0 = 1$ can be either *true* or *false*.
- In practice, we are interested in a specific class of interpretations.
- The specific logic, called **First-order theories**, is built by imposing restrictions on nonlogical symbols and their meanings.